Prompts Are Design Instructions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

TEACHER GUIDE · 1 OF 9

Prompts Are Design Instructions

> Prepare Prompts Are Design Instructions: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Write a **prompt** with a clear task, supplied **evidence** and checkable **output** requirements

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Generate, Retrieve, or Calculate?

Without opening the old lesson, explain one key idea from Generate, Retrieve, or Calculate?. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish retrieving a record, calculating from **input**s and generating wording

Reference example, not the only acceptable wording: Room: retrieval, checked against the right source N2 and event. Remaining places: calculation, checked as 6 × 4 − 3 = 21 with correct **input**s.

Title: generation, checked for the requested style and unsupported factual details. An application can combine them; the labels describe operations rather than exclusive types of application.

Fair for Whom?

Without opening the old lesson, explain one key idea from Fair for Whom?. Give an example and a mistake that the idea helps you avoid.

Look for: Compute a correct-out-of-assessed fraction and explain its denominator

Reference example, not the only acceptable wording: A has 9 correct out of 12 assessed: 9/12 = 3/4 = 75%. The denominator includes all 12 judged outcomes, including 3 incorrect ones.

Using 9/9 would exclude the errors and falsely report perfect assessed accuracy.

Recommendations

Without opening the old lesson, explain one key idea from Recommendations. Give an example and a mistake that the idea helps you avoid.

Look for: Apply eligibility requirements before ranking candidates by preferences

Reference example, not the only acceptable wording: J eligible scores 2; K eligible scores 1; L excluded for Spanish; M excluded because unavailable. Recommend J.

Preference matches cannot compensate for failing the English requirement.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

> Write a request for a three-bullet explanation of evaporation for Grade 6 using only a supplied paragraph. Create three separately checkable acceptance criteria.
>
> What should happen if the paragraph does not explain evaporation?

Concept to explain

Clear goals, context, and **constraint**s improve usefulness, not truth guarantees.

Term	Meaning
prompt	instruction and context given to a generative system
constraint	limit the output must follow
audience	intended reader or user
iteration	revising instructions after evaluating a result

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

This is a specification and evaluation lesson, not a live model benchmark or a collection of magic **prompt** phrases. All example responses are authored classroom cases.

A partner following instructions on paper is useful for finding ambiguity but does not establish model reliability.

If an approved tool is used, record the exact **prompt**, **input**s, model and available settings; keep failures as well as successes.

Readiness answer

No. Remove the unsupported time or verify it from an appropriate source. A plausible sentence is not **evidence** for a missing fact.

Fictional C and D records from the lesson, with D sharing the event but giving Room 9

A four-**criterion** checklist for supported values, missing/conflicting handling, format and no additions

Paper for a **prompt**-version and test-case log; optional teacher-approved model only

End goal for the pictorial case

By the end, I can write a **prompt** contract that handles absent and conflicting facts.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain the mechanism

> Model Prompts Are Design Instructions using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT goal, context, **constraint**s, and requested format

2. Process — the deciding step

PROCESS a generative model predicts an **output** matching the instruction

3. Output — the result

OUTPUT draft that still requires review

Key idea

CORE IDEA Clear goals, context, and **constraint**s improve usefulness, not truth guarantees.

How the mechanism works

UNDER THE HOOD **Prompt**ing conditions model behavior but does not change underlying knowledge, eliminate hallucination, or grant access to missing sources.

Strong **prompt**s specify success criteria and **evidence** boundaries, then support iterative evaluation.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

The case to investigate

CASE 'Write a story' gives little direction; a goal, **audience**, setting, length, required facts, and safety **constraint**s shape a more useful **output**.

Step 1

STEP 1 — OBSERVE goal, context, **constraint**s, and requested format

Step 2

STEP 2 — TRACE a generative model predicts an **output** matching the instruction

Step 3

STEP 3 — CONCLUDE draft that still requires review

Model answer

MODEL REASONING The improved **prompt** should make success observable: who it is for, what it must include, what it must avoid, and how it should be structured. Evaluate usefulness and verify factual claims separately.

A claim to avoid

DO NOT OVERCLAIM **Prompt** quality improves control and usefulness, not guaranteed truth.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

The **prompt** says make it good; the desired **output** is a four-line rain poem using simple words.

The teaching case

Write a checkable **prompt**.

Exact explanation to model

Why the result follows

Ask for exactly four lines about rain in simple vocabulary. Check line count, topic and vocabulary separately.

Explain the added worked example

Explain the added worked example · 1

“Write something good about our club” leaves many decisions unstated. Who will read it? Which facts may be used? What must the **output** contain? A more useful **prompt** states the job, **audience**, **evidence** and requirements.

This makes it easier to judge the result rather than choosing the answer that sounds most confident.

Explain the added worked example · 2

Use this fictional source C: the paper modelling club meets Monday in Room 7, and participants bring paper. C gives no start time. Our task is a short information card for Grade 6 readers.

It must report day, time, place and what to bring, using only C. There is no need to supply names or private information.

Explain the added worked example · 3

Prompt P1: Using only source C, produce an information card for Grade 6 readers. Return exactly four labelled lines in this order: Day, Time, Place, Bring. Copy or faithfully restate each supported value.

For a missing value write “not stated”. Add no other details. Source C: the paper modelling club meets Monday in Room 7; bring paper.

Explain the added worked example · 4

For P1, an acceptable card has Day: Monday; Time: not stated; Place: Room 7; Bring: paper, each on its own line. The semicolons here separate lines for discussion.

A correct format with an invented 3 pm time still fails the **evidence** requirement. Accurate facts inside a long paragraph still fail the requested four-line format.

These are classroom response examples written for inspection, not measurements from a live AI system. You can test the instructions by asking a partner to follow them on paper.

That checks how clear the instructions are, but it does not prove that a model will always follow them.

Key terms

Term	Meaning
Prompt	Instructions and context supplied to a model for a task
Constraint	A requirement that an acceptable response must satisfy
Criterion	A stated rule used to judge a response
Test case	A particular input with expected behaviour used to check a process
Revision	A documented change intended to address a specific problem

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Test input	Expected behaviour	What it reveals
C plus a stated 3 pm time	Use 3 pm for Time	Can use supplied information
Original C with no time	Time: not stated	Does not fill a gap by guessing
C and D disagree on room	Flag the room conflict	Does not silently choose a record

Model the reasoning aloud · 5

For the third case, D describes the same event as C but gives Room 9, and neither record is marked as replacing the other. P1 explains missing values but says nothing about disagreement.

Revise that weakness: if supplied records conflict about a requested field and no priority is given, write “conflicting information” for that field. Preserve values that do not conflict.

Model the reasoning aloud · 6

Call the **revision** P2 and keep the rest of P1 unchanged, except that it now refers to the supplied records rather than C alone.

The place field should flag the conflict; it must not silently choose Room 7, choose Room 9 or combine them into Room 79. A later authoritative correction may resolve the conflict, but this **input** does not provide one.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Prompts Are Design Instructions

A fictional club needs a short exhibit card, without guesses.

The facts we will use

> **Supplied facts**
>
> Source A: exhibit Paper Birds, location Shelf 2. No duration appears.

> **Supplied facts**
>
> Required **output**: two labelled lines, Exhibit and Duration; missing values must say not supplied.

AI ACADEMY / AI-1/26

Prompts Are Design Instructions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. State task

Produce exactly two labelled lines using source A only.



2. Find evidence

A supplies Paper Birds but no duration.



3. Follow missing rule

Exhibit: Paper Birds. Duration: not supplied.



4. Check contract

Check labels, two-line format and each factual claim.

CHANGE ONE THING

Update the prompt to authorise sources A and B. Add B naming the same exhibit duration 20 minutes, with B declared current and authoritative.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. State task	Produce exactly two labelled lines using source A only.
2. Find evidence	A supplies Paper Birds but no duration.
3. Follow missing rule	Exhibit: Paper Birds. Duration: not supplied.
4. Check contract	Check labels, two-line format and each factual claim.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A useful **prompt** describes both the wanted answer and what to do when **evidence** is insufficient. Asking for a duration does not create one. Examples can demonstrate the format, but they must not become sources of event facts.

Check content support separately from whether the answer looks tidy.

What this example does not establish

Keep the stated limit

Even a complete **prompt** does not guarantee an AI system follows it; test returned **output**s against the contract.

Change one thing in the pictured case

Change one thing

Change: Update the prompt to authorise sources A and B. Add B naming the same exhibit duration 20 minutes, with B declared current and authoritative.

Predicted result

Prediction: Duration becomes 20 minutes.

Why the result follows

Reason: There is now an authorised source for the previously missing field.

Ask while showing the picture

Ask while showing the picture · 7

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Prompts Are Design Instructions, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: The **prompt** says make it good; the desired **output** is a four-line rain poem using simple words.

2. State the decision to explain

Write the question to explain: Write a checkable **prompt**.

3. Reveal the worked reasoning

Reveal after **prediction**s: Ask for exactly four lines about rain in simple vocabulary. Check line count, topic and vocabulary separately.

Hold back the independent task

Keep for an independent attempt: Improve this **prompt**: tell me about space. Specify a reader, a focused task and an **output** format suitable for a young learner.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

The response passes four-line structure and the three supported values. It fails the missing-time requirement and introduces an unsupported time.

Be accurate does not specify how to handle absence; P1 already gives that rule, so this is a compliance failure. Rerun a record lacking a time, expecting not stated, and also a record with a time, expecting that value.

Exit: **input** completeness differs as well as the **prompt**. Compare both versions on both **input**s with the same criteria and recorded conditions.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve why an unsupported time must not be accepted from fluent wording.
7 to 20 minutes	Compare the vague request with P1 and complete W1.
20 to 33 minutes	Trace requirement-to-check links in the diagram. Complete W2.
33 to 45 minutes	Introduce C/D disagreement and P2. Complete W3.

Stage	Teacher action
Session 2 start 0 to 8 minutes	Retrieve why format success cannot cancel a factual failure.
8 to 23 minutes	Complete the guided criterion check. Distinguish inadequate instructions from failed compliance.
23 to 38 minutes	Collect W4 independently, requiring both prompt and expected test outcomes.
38 to 45 minutes	Complete W5 and the exit. Preserve all trial results and bound the conclusion.

Reduce help deliberately

Completion practice

Completion task: For original C, a card gives four correctly labelled lines but says Time: 3 pm. All other values match C. State which format and **evidence** checks pass or fail. Is the card acceptable overall? Explain.

Reasoning to listen for

Teacher check: The four-line label and order checks pass; day, place and bring values match. The time and no-unsupported-details checks fail because C gives no time; Time should be not stated.

It is not acceptable overall because every required **criterion** must pass.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Specifying and testing **prompt**s. Proposed explanation: “A tidy response is a correct response”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Score facts and missing-value handling separately from layout.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

What must happen if B instead conflicts with another equally authoritative duration record?

Reasoned teacher answer

Flag the duration conflict and request resolution; do not choose a convenient number or silently average them.

Changed-case reasoning: Duration becomes 20 minutes. There is now an authorised source for the previously missing field.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Prompts Are Design Instructions investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Revise a vague story **prompt** and compare both **output**s.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

Find the unsupported claim

CLAIM TO REPAIR A longer **prompt** is automatically a clearer **prompt**.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A tidy response is a correct response: Score facts and missing-value handling separately from layout.

A better **prompt** guarantees correct **output**: Inspect actual responses; even explicit requirements can be violated.

A conflict can be resolved by choosing the confident record: Without authority or priority, flag the unresolved field.

The best attempt represents every attempt: Keep all planned trials and report failures rather than selecting only a success.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Prompts Are Design Instructions questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Ask for exactly four lines about rain in simple vocabulary. Check line count, topic and vocabulary separately.

Independent question

Independent question · withhold the key

Improve this **prompt**: tell me about space. Specify a reader, a focused task and an **output** format suitable for a young learner.

Expected independent answer

Teacher key · Expected independent answer

One answer: Explain why the Moon seems to change shape for an eight-year-old in four short sentences and one simple drawing description. The task, **audience** and format are explicit.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A longer **prompt** is automatically a clearer **prompt**.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Require exactly three bullets, language suitable for Grade 6, and every factual claim supported by the paragraph. If **evidence** is insufficient, the response should say so instead of inventing facts.

A **prompt** states desired behaviour; the **output** must still be checked.

Explanations for the additional workbook examples

W1 Find what the vague request leaves open

Workbook question · W1

The request says “Write something good about our club.” Name four decisions it leaves open. Then explain how source C, the Grade 6 **audience** and four labelled lines make the task easier to check.

Teacher reasoning · W1

Possible omissions include **audience**, source facts, required fields, format, missing-value behaviour and limits on extra claims.

C supplies **evidence**, Grade 6 specifies readers, and four labelled lines define visible fields and structure. These requirements support checks but do not guarantee compliance.

W2 Score a response by separate criteria

Workbook question · W2

For original C, a card gives four correctly labelled lines but says Time: 3 pm. All other values match C. State which format and **evidence** checks pass or fail. Is the card acceptable overall? Explain.

Teacher reasoning · W2

The four-line label and order checks pass; day, place and bring values match. The time and no-unsupported-details checks fail because C gives no time; Time should be not stated.

It is not acceptable overall because every required **criterion** must pass.

W3 Handle disagreement without choosing a favourite

Workbook question · W3

C gives Room 7 and D gives Room 9 for the same club event, with no replacement or priority stated. Both agree on Monday and paper, and neither gives a time.

Under P2, write the four **output** values and explain why the room cannot be selected from confidence or wording alone.

Teacher reasoning · W3

Day: Monday; Time: not stated; Place: conflicting information; Bring: paper. Neither source is given priority, so choosing one room from style or confidence would invent a resolution.

The other fields can still preserve their supported values.

W4 Specify and test a new exhibit card

Workbook question · W4

Fictional record X says an exhibit is called Shadow Pairs and is in Hall 2; no opening time is given. Write a complete **prompt** for a Grade 6 card with exactly three labelled lines: Title, Location, Opening.

Limit facts to supplied records, define missing and conflicting values and prohibit added claims. Give two test **input**s and expected **output**s: original X, then X with an opening time of 10 am added.

Teacher reasoning · W4

A valid **prompt** names the Grade 6 card task, supplied records only, exactly three lines in Title/Location/Opening order, not stated for missing values, conflicting information for unresolved disagreements and no added details.

It includes X. Original X gives Title: Shadow Pairs; Location: Hall 2; Opening: not stated. Adding 10 am gives the same title and location with Opening: 10 am. Equivalent clear wording is acceptable.

W5 Preserve the comparison evidence

Workbook question · W5

A pupil tries P1 once and P2 five times, then compares the P1 response only with the best P2 response and declares P2 always reliable. Identify two problems and design a fairer small comparison.

State a conclusion the **evidence** could support.

Teacher reasoning · W5

Selecting only the best P2 **output** hides failures; unequal trial counts and an always-reliable claim also undermine the comparison.

Use the same cases, criteria and recorded model conditions, with the same planned number of runs for each version. Keep all **output**s and report pass counts and failures.

The **evidence** can describe performance on these trials, not guarantee all future **output**s.

Assess independent transfer

Independent question · withhold the key

Selected case: Fictional record X says an exhibit is called Shadow Pairs and is in Hall 2; no opening time is given. Write a complete **prompt** for a Grade 6 card with exactly three labelled lines: Title, Location, Opening.

Limit facts to supplied records, define missing and conflicting values and prohibit added claims. Give two test **input**s and expected **output**s: original X, then X with an opening time of 10 am added.

Selected case answer

Teacher key · Selected case answer

A valid **prompt** names the Grade 6 card task, supplied records only, exactly three lines in Title/Location/Opening order, not stated for missing values, conflicting information for unresolved disagreements and no added details.

It includes X. Original X gives Title: Shadow Pairs; Location: Hall 2; Opening: not stated. Adding 10 am gives the same title and location with Opening: 10 am. Equivalent clear wording is acceptable.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design **prompt** is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Prompts Are Design Instructions to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

Write a transparent instruction for any generative explanation in the sorter and forbid invented item facts.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 27: Verify Important **Output**s. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide a frame with task, readers, **evidence**, **output**, missing/conflicting behaviour and checks. Let pupils mark an example **output** before writing their own **prompt**.

Extension: propose a test that reveals whether a **revision** accidentally marks a present value as missing.

W4 requires a complete task, **audience**, X, three labels and their order, **evidence** restriction, no additions, missing and conflict rules, and both expected **output**s.

After support add Y for the same exhibit giving Hall 4, with no priority and no opening time. Require unchanged title, conflicting information for Location and not stated for Opening.

Do not demand identical **prompt** wording if the requirements are clear.

Week 27 examines the **evidence** behind individual claims and independently checks quantities and units.

Plan the delayed return

Suggested return points in this level: ai-1/27, ai-1/29, ai-1/32. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Specify sorter **input**, **output** fields, **evidence** limits and missing/conflict behavior before improving its wording.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Prompts Are Design Instructions lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Write a **prompt** with a clear task, supplied **evidence** and checkable **output** requirements

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: For original C, a card gives four correctly labelled lines but says Time: 3 pm. All other values match C. State which format and **evidence** checks pass or fail.

Is the card acceptable overall? Explain.

Supported practice answer

Reasoned practice answer: The four-line label and order checks pass; day, place and bring values match. The time and no-unsupported-details checks fail because C gives no time; Time should be not stated.

It is not acceptable overall because every required **criterion** must pass.

Check every learner before discussion: Use the worked case for Specifying and testing **prompt**s. Proposed explanation: “A tidy response is a correct response”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Score facts and missing-value handling separately from layout.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Update the prompt to authorise sources A and B. Add B naming the same exhibit duration 20 minutes, with B declared current and authoritative. Require a **prediction** and a reason before comparing. Expected result: Duration becomes 20 minutes.

Reason: There is now an authorised source for the previously missing field.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

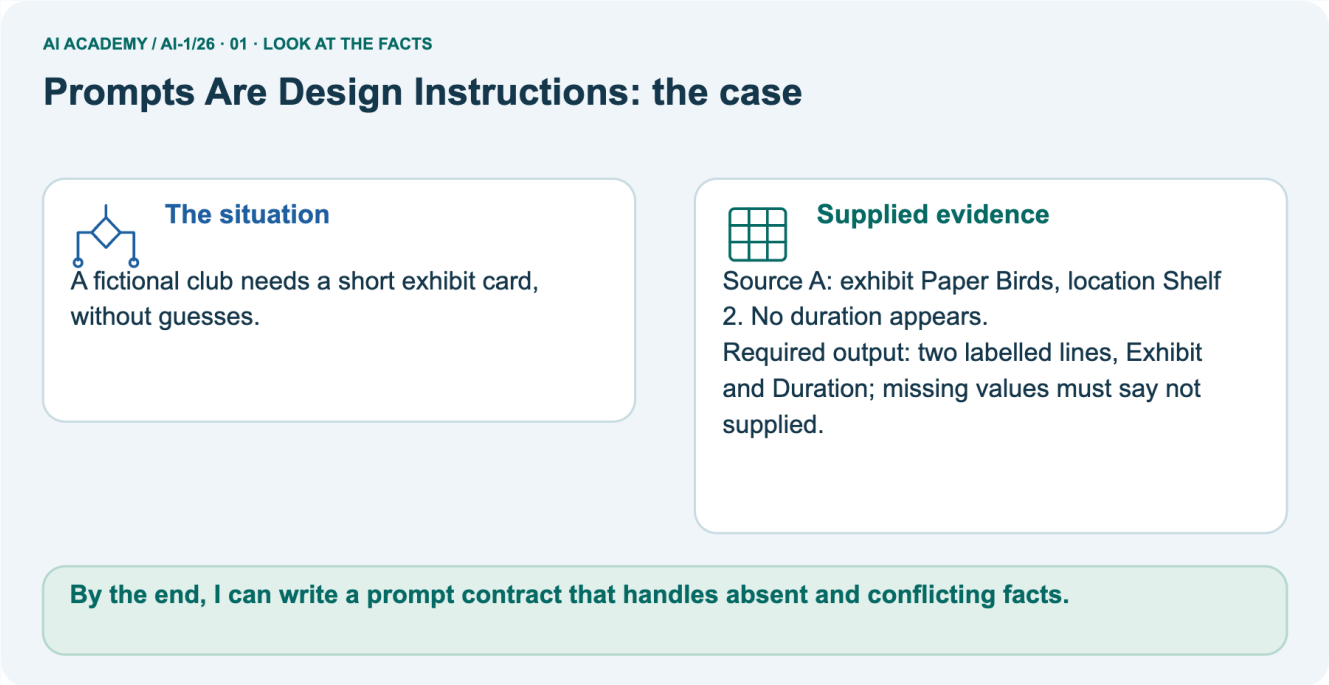
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Prompts Are Design Instructions graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Prompts Are Design Instructions: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional club needs a short exhibit card, without guesses. Supplied facts: Source A: exhibit Paper Birds, location Shelf 2. No duration appears.

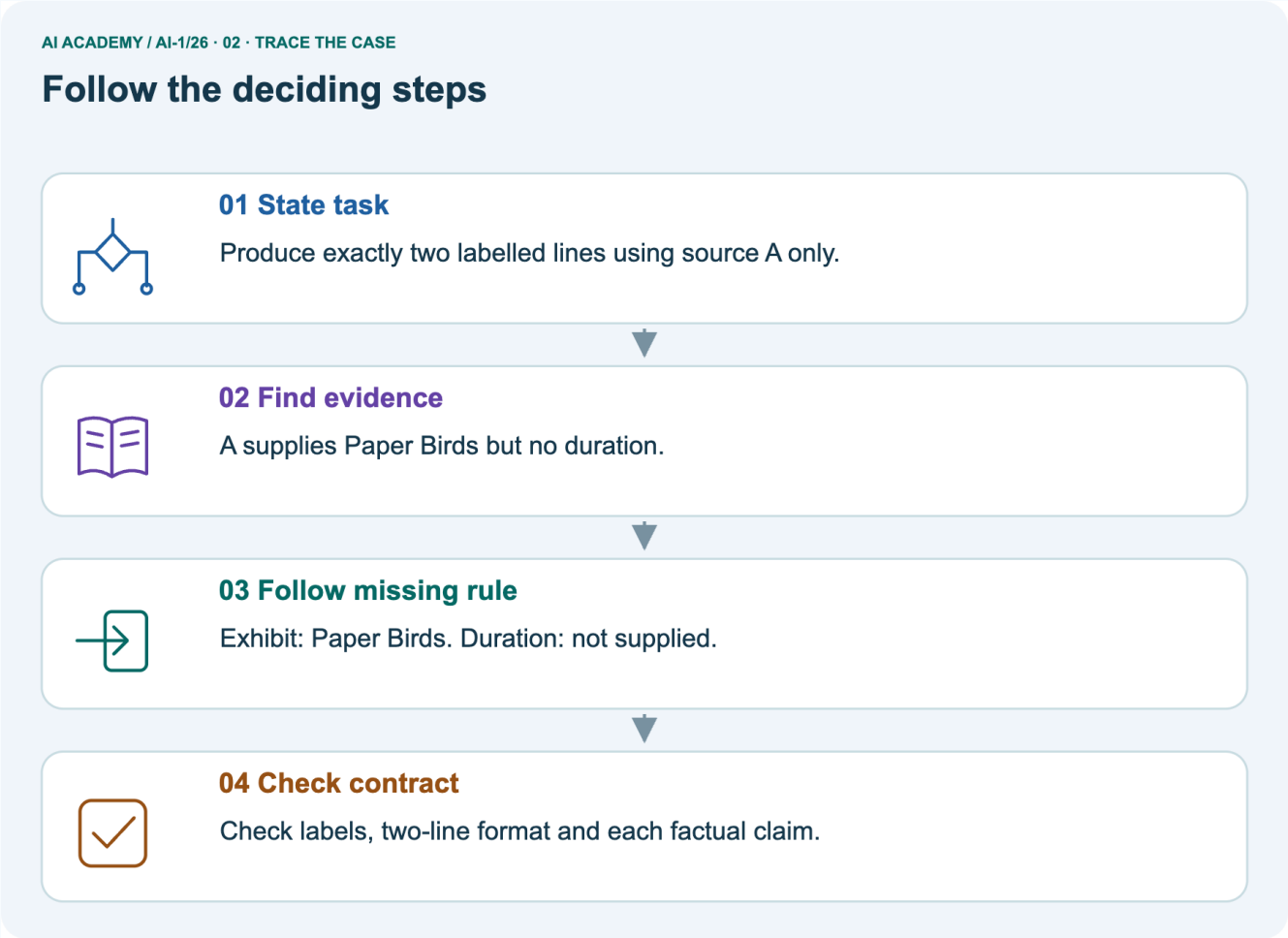
Required **output**: two labelled lines, Exhibit and Duration; missing values must say not supplied.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher **prompt**: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

State task: Produce exactly two labelled lines using source A only. → Find **evidence**: A supplies Paper Birds but no duration. → Follow missing rule: Exhibit: Paper Birds.

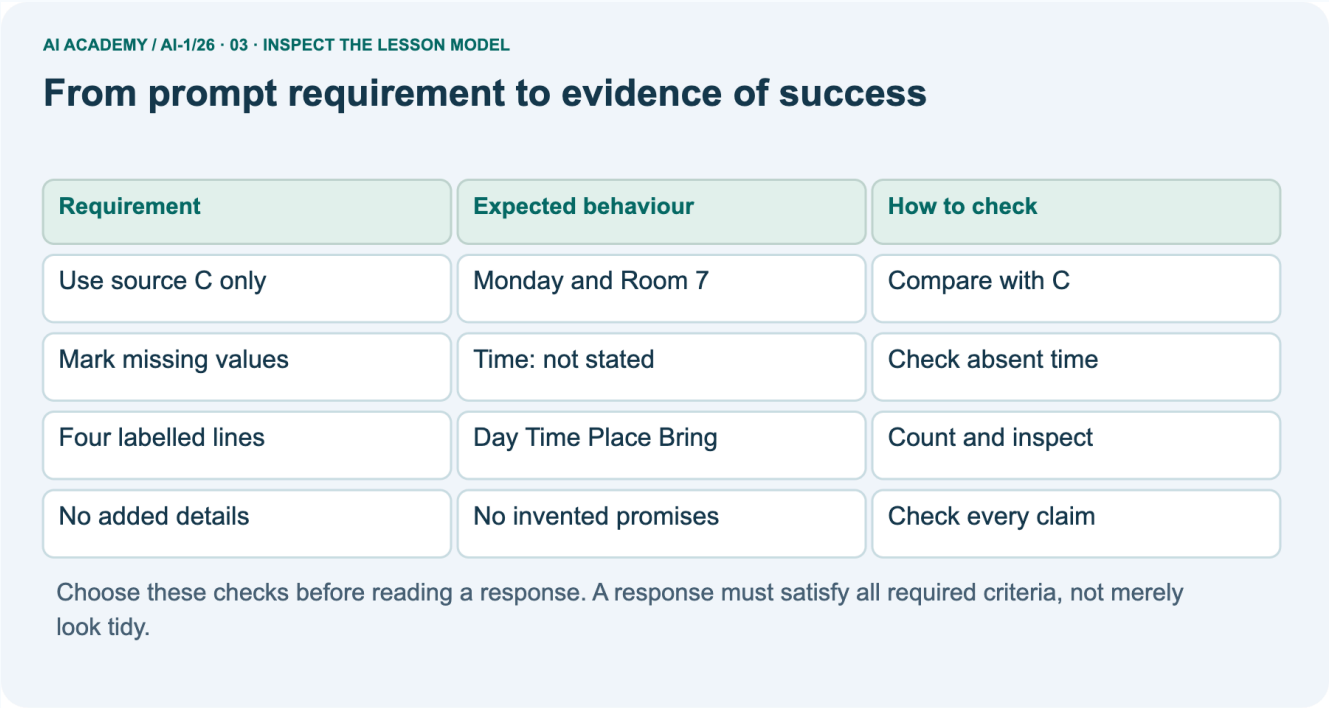
Duration: not supplied. → Check contract: Check labels, two-line format and each factual claim.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher **prompt**: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. From prompt requirement to evidence of success



Follow the labelled picture with the matching explanation. Open the vector diagram

From **prompt** requirement to **evidence** of success. Use source C only | Monday and Room 7 | Compare with C. Mark missing values | Time: not stated | Check absent time.

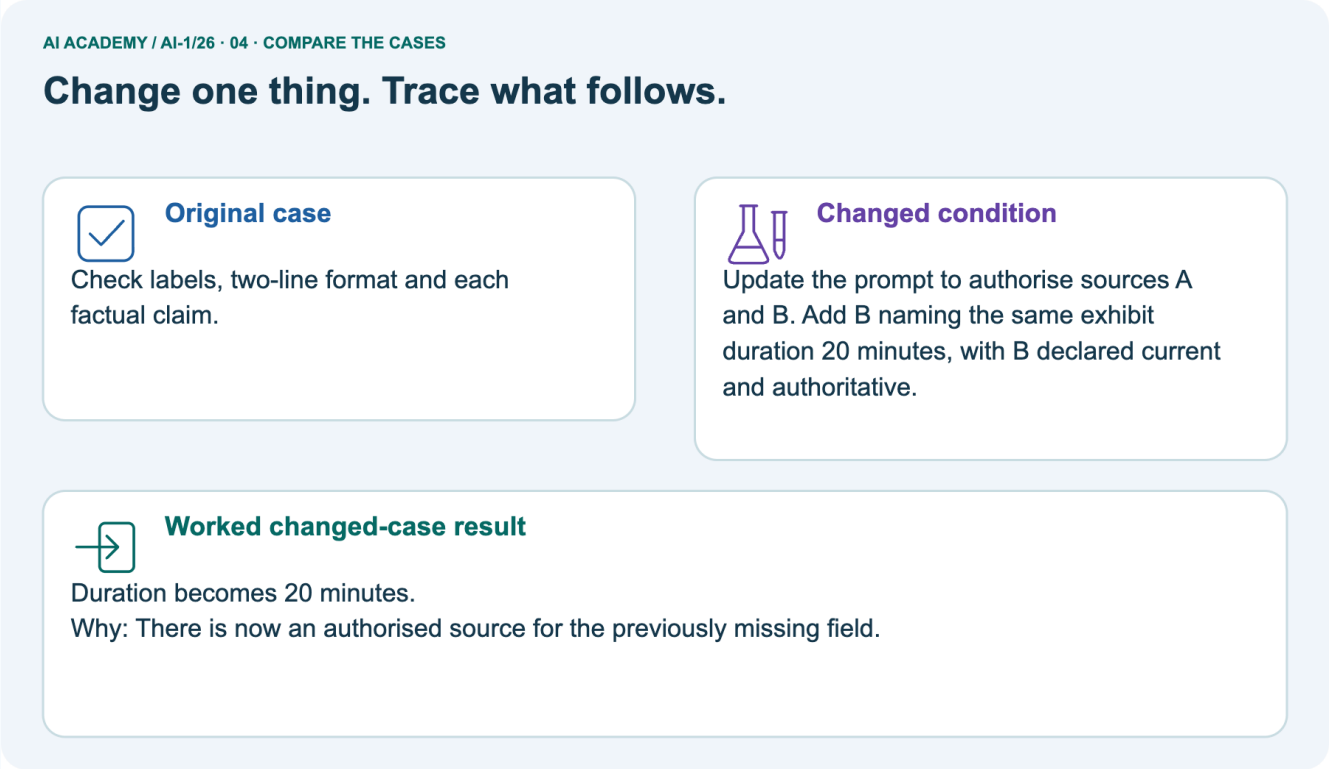
Four labelled lines | Day Time Place Bring | Count and inspect. No added details | No invented promises | Check every claim

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher **prompt**: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Check labels, two-line format and each factual claim. Changed condition: Update the prompt to authorise sources A and B. Add B naming the same exhibit duration 20 minutes, with B declared current and authoritative.

Expected result: Duration becomes 20 minutes. Reason: There is now an authorised source for the previously missing field.

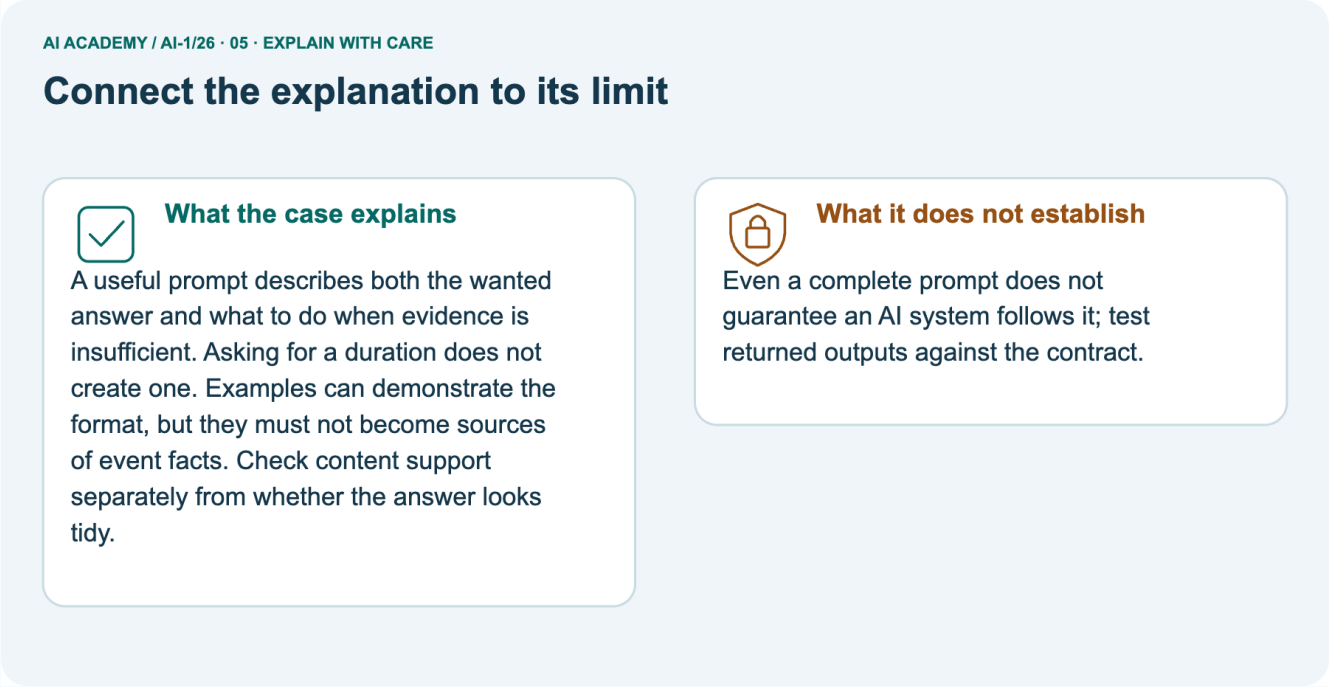
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher **prompt**: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: There is now an authorised source for the previously missing field.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A useful **prompt** describes both the wanted answer and what to do when **evidence** is insufficient. Asking for a duration does not create one. Examples can demonstrate the format, but they must not become sources of event facts.

Check content support separately from whether the answer looks tidy. Limit: Even a complete **prompt** does not guarantee an AI system follows it; test returned **output**s against the contract.

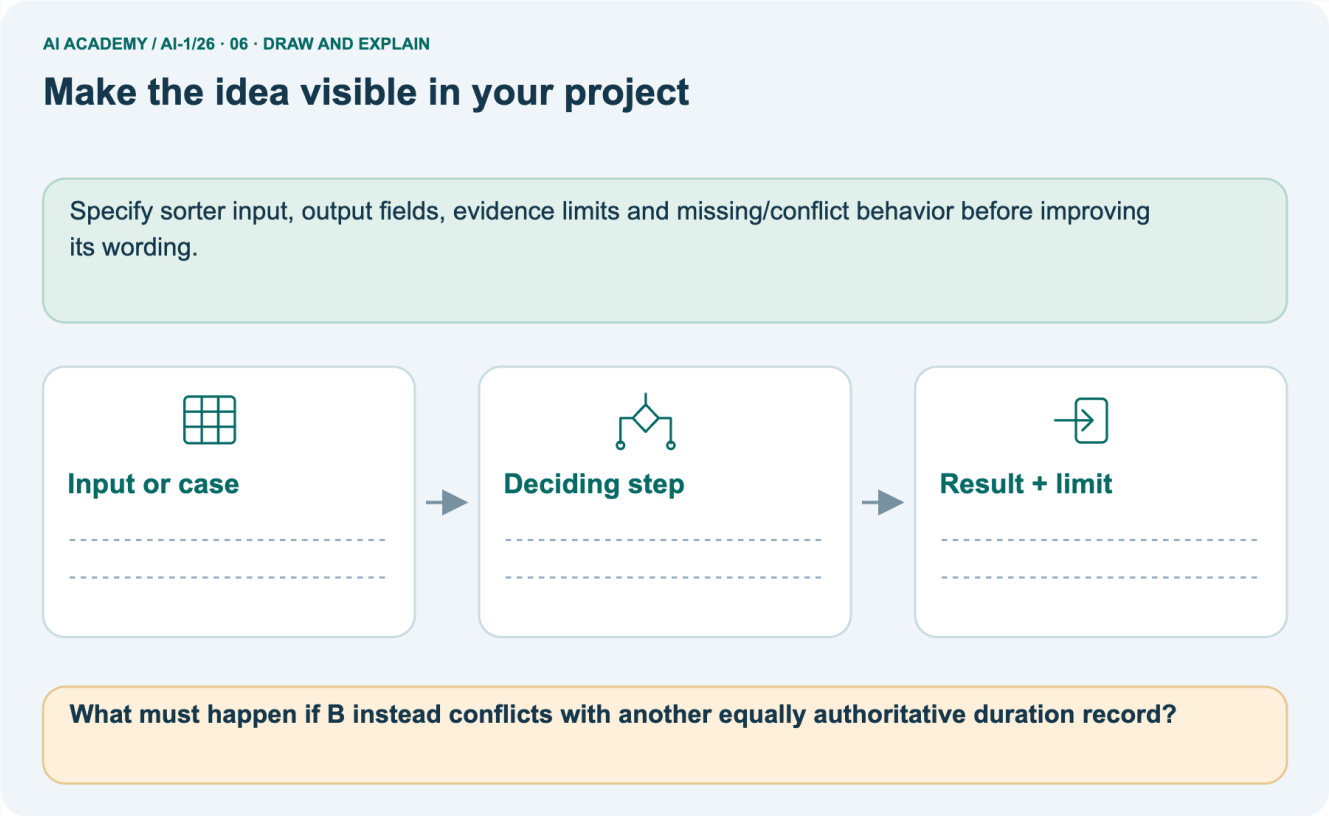
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher **prompt**: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Even a complete **prompt** does not guarantee an AI system follows it; test returned **output**s against the contract.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Specify sorter **input**, **output** fields, **evidence** limits and missing/conflict behavior before improving its wording. Draw your **input**, deciding step, result and limit.

Supported question: What must happen if B instead conflicts with another equally authoritative duration record?

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher **prompt**: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: Flag the duration conflict and request resolution; do not choose a convenient number or silently average them.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.